SUMMARY (cont.)

- Stokes formula F= 6πηav, where F is the viscous force acting on a sphere of radius a and v is the terminal velocity of the sphere.
- The surface tension of a liquid is defined as the force of tension acting on a unit length of an imaginary line drawn on the free surface of the liquid, the direction of the force being perpendicular to the line so drawn and acting parallel to the surface.
- The angle between tangents drawn at the point of contact to the liquid surface and solid surface inside the liquid is called the *angle of contact* for a pair of solid and liquid.
- The flow of a fluid is said to be steady if, at any given point, the velocity of each passing fluid particle remains constant with respect to time.
- The equation $a_1 v_1 = a_2 v_2$ is called the equation of continuity for a flow of fluid through a tube and it is due to the conservation of mass in the flow of fluids. It states that the sum of pressure energy, kinetic energy, and potential energy per unit mass of an incompressible, non-viscous fluid in a streamlined flow remains constant. i.e., $P/\rho + v^2/2 + gh = \text{constant}$.

UNIT-7(XI-Physics_Vol-2)_14-08-2020.indd 88 17-08-2020 11:31:27